Zain Dana Harper

Systems and verification engineer. Accountability infrastructure for AI agents, compilers and typed effects, real-time graphics and color.

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Summary

I build systems that someone who was not in the room can check. That means capability gates a model cannot talk past, receipts that re-derive offline from the record alone, and a compiler where reaching for the network is part of a function's type.

Public engineering practice since 2023, on about thirteen years of programming and eleven years running field operations. The same discipline covers the rest of it: real-time graphics and color science, technical and compliance documentation, and research systems that carry their evidence with them. Every claim below opens onto something public.

Selected engineering

Flywheel, accountability engine for AI agents

Python, stdlib-only engine, with a Flutter client. The public flagship.

- Every tool call a model makes passes a default-deny capability gate and emits a sealed, hash-chained receipt binding what was permitted to what happened. Receipts witness argument and output digests, never raw content, and a third party re-verifies the chain offline.
- Receipts compose into a witness graph, so one drifted action degrades exactly its downstream dependents and nothing further.
- A domain oracle registry routes each claim to an independent verifier: pytest for code, the Lean kernel for mathematics, a measurement gate for empirical claims. Where no verifier exists the answer is UNVERIFIABLE, which is the honest one.
- Shipped an infrastructure control package covering network egress receipts, credential scanning, isolation acceptance tests, a dual-confirmation kill switch, and cross-layer event correlation, with a dated assessment mapping it against the failure classes of the July 2026 agentic-security incidents. Gaps are stated per class.
- Ran the engine's first preregistered confirmatory study in August 2026. Across 2,646 certificate bodies in nine model contexts, zero verdict disagreements, so acceptance is a function of the certificate and not of the model. Held-out selection beat random selection at every model size on the solvable task family. The harder family defeated every model on the ladder, and that null is published as a null.

BuildLang, compiler and language tooling

Rust, C, HLSL. Public crate surface.

- A systems language where ambient access belongs to a function's type. Filesystem, network, and foreign-call effects are tracked through closures, struct fields, control flow, and async blocks, so a callback cannot quietly launder authority it was never granted.
- Hindley-Milner inference, sum types, two-way C FFI, an experimental linear-types attribute with honestly scoped soundness, and a production C backend that also emits HLSL and GLSL.
- The buildc toolchain covers build, run, test, repl, fmt, pkg, doctor, and an LSP server. A checked build writes a receipt, and buildc receipt verify re-derives it later.

Project Telos, tools for model-era engineering

Python, TypeScript, Node. Versioned public packages.

- Fourteen public engines. Each one stands alone, and each can hand verified output to the next: repository maps and context packs, research intake that carries provenance, model-agnostic orchestration on a replayable ledger, worker and verifier evaluation, byte-level integrity witnessing, accountable learning, and agent memory that keeps provenance through recall.

Real-time graphics, color, and native systems

C++, HLSL, Python.

- Built D3D11 and HLSL rendering through proxy-DLL interception and mid-frame compute dispatch: ACES and AgX tone mapping, TAA, SSR, SSGI, GTAO, SDSM, volumetrics, and a read-only shared-memory bridge carrying live game state to the GPU.
- Elder ENB, a graphics and lighting preset for Skyrim Special Edition, has passed 930,000 downloads across roughly 280 releases over two years. Shipping to an active user base that long is where the release discipline came from.
- Color work covers perceptual spaces, color difference, HDR tone mapping, gamut tools, ICC and VCGT profiles, and display characterization.

Research

Eight public records with permanent DOIs: systems papers, preprints, research notes, and archived corpora. None of it is peer reviewed and the pages say so. The full list with exact status sits on the CV, and the program it belongs to is in the dossier.

- **EMET: An Authority-Incapable Byte-Level Integrity Witness.** Systems paper with a public implementation. It re-derives byte-integrity facts and refuses to emit trust, safety, or permission. DOI (https://doi.org/10.5281/zenodo.21230267)
- **Witnessed Independence: Recording Whether a Verifier Graded Its Own Work.** Published preprint. DOI (https://doi.org/10.5281/zenodo.21232206)
- **Proof Packets: A Derive-Don't-Trust Envelope for Accountable Agent Actions.** Published preprint. DOI (https://doi.org/10.5281/zenodo.21231406)
- **Conferred Existence.** Long-form thesis on made minds, perception, memory, and standing. Archived research corpus. DOI (https://doi.org/10.5281/zenodo.20773724)

Open source

Work sent into repositories other people own, where a maintainer sets the standard and makes the call. Counted from the GitHub API by tools/pr_census.py, and every entry opens onto the diff.

23 merged | **20** repositories | **12** open | **11** closed without merge

- Merged work covers defect repair, missing validation, and test coverage in Datasette, tomlkit, pydantic-ai, DeepEval, pydash, and grimp. Examples: the correct HTTP status for a malformed composite-key row URL, a TOML round-trip bug where an array of tables swallowed the next sibling key, and a metric that crashed on unhashable tool outputs.
- A further **14** pull requests add one row naming a tool to a public link index. They change no software, so they are held apart from the counts above rather than folded into them.
- Open is not accepted, and the misses stay on the record. The full ledger, including every declined pull request, is on the portfolio.

Speaking and service

- **Puget Sound Programming Python.** "Pick the Lock for Everyone: Building Verifiable AI Workflows in Python," invited talk, scheduled 2026-08-19.
- **Coalition for Secure AI.** Individual contributor to Workstream 4, Secure Design for Agentic Systems, contributing secure-design patterns and a reference implementation.
- **PlantAmnesty.** Volunteer time every year since 2015 with a Seattle network of horticulturists and arborists, supporting workshops and member events. Self-reported; a service-history letter has been requested.

Experience

Independent systems engineer and founder · Zentropy Labs

2023 TO NOW · SEATTLE AND REMOTE

- Own architecture, implementation, tests, documentation, packaging, demos, and release evidence across a multi-repository public portfolio.
- Run coordinated coding agents for parallel discovery and implementation, and keep first-party review, verifier separation, and public claim discipline explicit.

Freelance technical writer, documentation and product operations

2017 TO NOW · REMOTE

- Write API and implementation guides, security and compliance documentation, proposals, release notes, and support material.
- Work with NIST 800-171, CMMC readiness, SOC 2, ISO 27001, DFARS, incident response, and audit support in a technical-writing capacity. Client names and deliverables stay out of the public portfolio.

Operations and commercial arboriculture lead · family business

2015 TO NOW · SEATTLE AREA

- Eleven years running tree crews from the ground. Rigged the loads, judged clearance and how far a limb would swing relative to structures and people, and served as the second set of eyes for the person in the air.
- Led client intake, estimates, site assessment, scheduling, safety calls, and crew and vendor coordination.

Technical networking support, Xbox division · Microsoft

2013 TO 2014 · REDMOND, WASHINGTON

- Diagnosed TCP/IP, DNS, NAT, firewall, router, and account-adjacent console networking faults over phone and chat, and wrote up the repeatable resolutions.

Strengths

LANGUAGES	Python, Rust, C++, TypeScript and JavaScript, Lua, HLSL, C#, PowerShell, Bash
VERIFICATION	Capability gates, sealed receipts, hash chains, offline re-verification, witness graphs, provenance, preregistration, evidence contracts
AI SYSTEMS	Model-neutral routing with quota failover, tool-use loops, MCP surfaces, multi-agent orchestration, worker and verifier separation, evaluation workflows
SYSTEMS	Compilers, type systems, typed effects, code generation, D3D11 and shader pipelines, shared-memory IPC, CMake and vcpkg
DELIVERY	Git and GitHub, pytest, GitHub Actions, Linux, package metadata, CLI design, release notes, developer documentation

Open the work

The fastest way to judge any of this is to open it. The portfolio holds the ledger and the samples, research holds the arguments and the experiments, the Studio runs in your browser right now, and github.com/HarperZ9 (https://github.com/HarperZ9) has the source and the releases.

Download this page: Resume PDF. The PDF is printed from this page, so the two cannot drift apart.

Updated 2026-08-13. Open-source counts are read from the GitHub API by tools/pr_census.py and gated by a test, so a number here that disagrees with the live record breaks the build.